

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures Assessment Tool 2 – Debugging & Optimisation Task 100
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	
Weightage:	35% of CIA	Total Marks:	
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
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Section / Batch:	3 rd cs&bs	Date:	24/07/2026

Code of Conduct

I _____ Sasikumar Megha _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____ Sasikumar Megha _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edgecase errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors

Code Quality & Readability	20	Clean, modular, welldocumented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand No or incorrect explanation No testing provided
Explanation & Justification	20	Clear, logical, and welljustified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____ Date: _____		Signature: _____ Date: _____		Signature: _____ Date: _____	

QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack - Balancing Symbols
5 marks



Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while(low < high) {
        int mid = (low + high) / 2;
        if(arr[mid] == key)
            return mid;
        else if(arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int binarysearch(int arr[],int n,int key){
3     int low=0,high=n-1;
4     while (low<=high) {
5         int mid=(low+high)/2;
6         if(arr[mid]==key)
7             return mid;
8         else if(arr[mid]<key)
9             low=mid+1;
10        else
11            high=mid-1;
12    }
13    return -1;
14 }
15 int main(){
16     int arr[]={2,4,6,8,12,15};
17     int n=sizeof(arr)/sizeof(arr[0]);
18     int key;
19     printf("enter element to search:");
20     scanf("%d",&key);
21     int result=binarysearch(arr,n,key);
22     if(result==1)
23         printf("element not found");
24     else
25         printf("element is found at index%d",result);
26     return 0;
27 }
```

📌 Submitted

Test Input

8

Output

enter element to search:element is found at index3

⏱ Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

🔍 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5];
3 int top = -1;
4 int pop() {
5     if (top == -1) {
6         printf("Underflow\n");
7         return -1;
8     }
9     return stack[top--];
10 }
11
12 void push(int val) {
13     if (top == 4) {
14         printf("Overflow\n");
15     } else {
16         stack[++top] = val;
17         printf("Pushed %d\n", val);
18     }
19 }
20
21 int main() {
22     push(10);
23     push(20);
24     printf("Deleted element: %d\n", pop());
25     return 0;
26 }
```

📌 Submitted

Test Input

stdin input (optional)

Output

```
Pushed 10
Pushed 20
Deleted element: 20
```

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

5/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int stack[5];  
3 int top;  
4  
5 void init() {  
6     top = -1;  
7 }  
8  
9 int main(void) {  
10     init();  
11     printf("Stack initialised\n");  
12     printf("Top = %d\n", top);  
13     return 0;  
14 }
```

Test Input

stdin input (optional)

Output

```
Stack initialised  
Top = -1
```

Pending manual review

Submitted

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```

if(ch == '}') {
    while(stack[top] != '{') {
        postfix[j++] = pop();
    }
    pop();
}

```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4
5 #define MAX 100
6
7 char stack[MAX];
8 int top = -1;
9
10 void push(char c) {
11     stack[++top] = c;
12 }
13
14 char pop() {
15     return stack[top--];
16 }
17
18 char peek() {
19     return stack[top];
20 }
21
22 int precedence(char c) {
23     if (c == '^') return 3;
24     if (c == '*' || c == '/') return 2;
25     if (c == '+' || c == '-') return 1;
26     return 0;
27 }
28
29 void infixto postfix(char infix[], char postfix[]) {
30     int i = 0, j = 0;
31     char ch;
32
33     while ((ch = infix[i]) != '\0') {
34         if (isalnum(ch)) {
35             postfix[j++] = ch;
36         } else if (ch == '(') {
37             push(ch);
38         } else if (ch == ')') {
39             while (top != -1 && peek() != '(') {
40                 postfix[j++] = pop();
41             }
42             if (top != -1 && peek() == '(') {
43                 pop();
44             }
45         } else {
46             while (top != -1 && precedence(peek()) >= precedence(ch)) {
47                 postfix[j++] = pop();
48             }
49             push(ch);
50         }
51         i++;
52     }
53
54     while (top != -1) {
55         postfix[j++] = pop();
56     }
57
58     postfix[j] = '\0';
59 }
60
61 int main() {
62     char infix[MAX], postfix[MAX];
63
64     scanf("%s", infix);
65
66     infixto postfix(infix, postfix);
67     printf("%s", postfix);
68
69     return 0;
70 }

```

Test Input

a+b*c

Output

abc*+

⚠ Pending manual review

📌 Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```

if(ch == '[' && stack[top] == '[') {
    (ch == '[' && stack[top] == '[') {
        pop();
    } else {
        return 0;
    }
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX 1000
5
6 int isbalanced(const char *expr) {
7     char stack[MAX];
8     int top = -1;
9
10    for (int i = 0; expr[i] != '\0'; i++) {
11        char ch = expr[i];
12
13        if (ch == '[' || ch == '{' || ch == '(') {
14            if (top < MAX - 1) {
15                stack[++top] = ch;
16            } else {
17                return 0;
18            }
19        }
20        else if (ch == ']' || ch == '}' || ch == ')') {
21            if (top > -1) {
22                return 0;
23            }
24        }
25        if ((ch == ')') && stack[top] == '(') ||
26            (ch == '}') && stack[top] == '{') ||
27            (ch == ']') && stack[top] == '[') {
28            top--;
29        } else {
30            return 0;
31        }
32    }
33    return (top > -1);
34 }
35
36 int main() {
37     char expr[MAX];
38
39     if (fgets(expr, sizeof(expr), stdin) == NULL) {
40         return 0;
41     }
42
43     expr[strlen(expr) - 1] = '\0';
44
45     if (isbalanced(expr)) {
46         printf("1");
47     } else {
48         printf("0");
49     }
50
51     return 0;
52 }

```

Submitted

Test Input

{(())}

Output

1

Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head;
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

🔧 **Debugging Task:** find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node* createnode(int data) {
10     struct node *newnode = (struct node*)malloc(sizeof(struct node));
11     newnode->data = data;
12     newnode->next = NULL;
13     return newnode;
14 }
15
16 void append(struct node **head, int data) {
17     struct node *newnode = createnode(data);
18
19     if (*head == NULL) {
20         *head = newnode;
21         return;
22     }
23
24     struct node *temp = *head;
25     while (temp->next != NULL) {
26         temp = temp->next;
27     }
28     temp->next = newnode;
29 }
30
31 void display(struct node *head) {
32     struct node *temp = head;
33
34     while (temp != NULL) {
35         printf("%d ", temp->data);
36         temp = temp->next;
37     }
38     printf("\n");
39 }
40
41 int main() {
42     struct node *head = NULL;
43
44     append(&head, 10);
45     append(&head, 20);
46     append(&head, 30);
47     append(&head, 40);
48
49     printf("linked list: ");
50     display(head);
51
52     return 0;
53 }
```

Test Input

stdin input (optional)

Output

linked list: 10 20 30 40

🔒 Pending manual review

📌 Submitted

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
PDP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define max 100  
3 int stack[max];  
4 int top = -1;  
5 void push(int item) {  
6     if (top == max - 1) {  
7         printf("Overflow\n");  
8     } else {  
9         stack[++top] = item;  
10    }  
11 }  
12  
13 int pop() {  
14     if (top == -1) {  
15         printf("Underflow\n");  
16         return -1;  
17     } else {  
18         return stack[top--];  
19     }  
20 }  
21  
22 int main() {  
23     int choice, x, item;  
24  
25     while (scanf("%d", &choice) != 1) {  
26         if (choice == 1) {  
27             if (scanf("%d", &x) != 1) break;  
28             push(x);  
29         } else if (choice == 2) {  
30             item = pop();  
31             if (item != -1) {  
32                 printf("%d\n", item);  
33             }  
34         } else if (choice == 3) {  
35             break;  
36         }  
37     }  
38  
39     return 0;  
40 }
```

Submitted

Test Input

```
1  
23  
1  
4  
-
```

Output

```
56  
4  
23
```

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

🔧 Debugging Task: find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node* createnode(int data) {
10     struct node *newnode = (struct node*)malloc(sizeof(struct node));
11     newnode->data = data;
12     newnode->next = NULL;
13     return newnode;
14 }
15
16 void insertend(struct node **head, int data) {
17     struct node *newnode = createnode(data);
18
19     if (*head == NULL) {
20         *head = newnode;
21         return;
22     }
23
24     struct node *temp = *head;
25     while (temp->next != NULL) {
26         temp = temp->next;
27     }
28     temp->next = newnode;
29 }
30
31 struct node* reverselist(struct node *head) {
32     struct node *prev = NULL;
33     struct node *current = head;
34     struct node *next = NULL;
35
36     while (current != NULL) {
37         next = current->next;
38         current->next = prev;
39         prev = current;
40         current = next;
41     }
42
43     return prev;
44 }
45
46 void printlist(struct node *head) {
47     struct node *temp = head;
48     while (temp != NULL) {
49         printf("%d ", temp->data);
50         temp = temp->next;
51     }
52 }
53
54 void freelist(struct node *head) {
55     struct node *temp;
56     while (head != NULL) {
57         temp = head;
58         head = head->next;
59         free(temp);
60     }
61 }
62
63 int main() {
64     struct node *head = NULL;
65     int choice, data;
66
67     while (scanf("%d", &choice) == 1) {
68         if (choice == 1) {
69             if (scanf("%d", &data) == 1) {
70                 insertend(&head, data);
71             }
72         } else if (choice == 3) {
73             printlist(head);
74             printf("\n");
75         } else if (choice == 4) {
76             head = reverselist(head);
77         }
78     }
79
80     freelist(head);
81     return 0;
82 }
```

Test Input

```
1
10
30
3

```

Output

```
10
10
10

```

🔴 Pending manual review

🟢 Submitted

QUESTIONS

Q1
Binary Search
5 marks



Q2
Stack
5 marks



Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Stack - Balancing Symbols
5 marks



Q6
Singly Linked List
5 marks



Q7
Stack
5 marks



Q8
Linked List
5 marks



Q9
Linked List
5 marks



Q10
Queue
5 marks



10/10 answered

Linked List

5 marks

What is the error in the below code?

```
new->next = head;
head = new;
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 struct Node *head = NULL;
10
11 void insertAtBeginning(int value) {
12     struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
13     if (newNode == NULL) {
14         printf("Memory allocation failed\n");
15         return;
16     }
17
18     newNode->data = value;
19     newNode->next = head;
20     head = newNode;
21 }
22
23 void display() {
24     struct Node *temp = head;
25     while (temp != NULL) {
26         printf("%d ", temp->data);
27         temp = temp->next;
28     }
29     printf("\n");
30 }
31
32 int main() {
33     int n, value;
34
35     scanf("%d", &n);
36     for (int i = 0; i < n; i++) {
37         scanf("%d", &value);
38         insertAtBeginning(value);
39     }
40
41     display();
42     return 0;
43 }
```

Submitted

Test Input

5
10 20 30 40 50



Output

50 40 30 20 10

Pending manual review

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

DEQUEUE(Q):

if front == -1:

print("Underflow")

else:

return Q[front + 1]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int Q[MAX];
6 int front = -1, rear = -1;
7
8 int dequeue() {
9     if (front == -1 || front > rear) {
10         printf("Underflow\n");
11         return -1;
12     }
13
14     int item = Q[front];
15     front++;
16
17     if (front > rear) { // queue became empty
18         front = rear = -1;
19     }
20
21     return item;
22 }
23
24 int main() {
25     int n, i, x;
26
27     scanf("%d", &n);
28
29     if (n <= 0) {
30         printf("Underflow\n");
31         return 0;
32     }
33
34     front = 0;
35     rear = n - 1;
36
37     for (i = 0; i < n; i++) {
38         scanf("%d", &Q[i]);
39     }
40
41     x = dequeue();
42
43     if (x != -1) {
44         printf("%d\n", x);
45     }
46
47     return 0;
48 }
```

Test Input

stdin input (optional)

Output

SET 2

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

```
PEEK(stack):  
return stack[top+1]
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 void push(int x) {  
9     if (top < MAX - 1) {  
10         stack[++top] = x;  
11     }  
12 }  
13  
14 int peek() {  
15     if (top == -1) {  
16         printf("Underflow\n");  
17         return -1;  
18     }  
19     return stack[top];  
20 }  
21  
22 int main() {  
23     int n, x;  
24  
25     scanf("%d", &n);  
26  
27     for (int i = 0; i < n; i++) {  
28         scanf("%d", &x);  
29         push(x);  
30     }  
31  
32     x = peek();  
33     if (x != -1) {  
34         printf("%d\n", x);  
35     }  
36  
37     return 0;  
38 }
```

Test Input

```
5  
10  
20  
30  
40
```



Output

50

🕒 Pending manual review

📌 Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 int isFull() {
9     return ((rear + 1) % MAX == front);
10 }
11
12 int isEmpty() {
13     return (front == -1);
14 }
15
16 void enqueue(int vehicle) {
17     if (isFull()) {
18         printf("Queue full\n");
19         return;
20     }
21
22     if (isEmpty()) {
23         front = rear = 0;
24     } else {
25         rear = (rear + 1) % MAX;
26     }
27
28     queue[rear] = vehicle;
29 }
30
31 int dequeue() {
32     if (isEmpty()) {
33         printf("Queue empty\n");
34         return -1;
35     }
36
37     int item = queue[front];
38
39     if (front == rear) {
40         front = rear = -1;
41     } else {
42         front = (front + 1) % MAX;
43     }
44
45     return item;
46 }
47
48 int main() {
49     int n, vehicle;
50
51     scanf("%d", &n);
52
53     for (int i = 0; i < n; i++) {
54         scanf("%d", &vehicle);
55         enqueue(vehicle);
56     }
57
58     while (!isEmpty()) {
59         printf("%d ", dequeue());
60     }
61
62     return 0;
63 }
```

Test Input

5 10 20 30 40 50

Output

10 20 30 40 50

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Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error.

if top == MAX:

Overflow

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value) {
9     if (top >= MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13     stack[++top] = value;
14 }
15
16 int main() {
17     int n, value;
18
19     scanf("%d", &n);
20
21     for (int i = 0; i < n; i++) {
22         scanf("%d", &value);
23         push(value);
24     }
25     printf("stack: ");
26     for (int i = top; i >= 0; i--) {
27         printf("%d", stack[i]);
28     }
29     printf("\n");
30     return 0;
31 }
```

Test Input

5

10

20

30

...

Output

stack: 504030010

🔍 Pending manual review

🟢 Submitted

QUESTIONS

Q1

Stack

5 marks



Q2

Circular Queue

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Doubly Linked List

5 marks



Q6

Queue

5 marks



Q7

Queue - Deque

5 marks



Q8

Binary Search

5 marks



Q9

Circular Queue

5 marks



Q10

Stack - Postfix Expression

5 marks



10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13     stack[++top] = value;
14 }
15
16 int pop() {
17     if (top == -1) {
18         printf("Underflow\n");
19         return -1;
20     }
21     return stack[top--];
22 }
23
24 int main() {
25     int n, value, deleted;
26
27     scanf("%d", &n);
28
29     for (int i = 0; i < n; i++) {
30         scanf("%d", &value);
31         push(value);
32     }
33
34     deleted = pop();
35     if (deleted != -1) {
36         printf("%d\n", deleted);
37     }
38
39     return 0;
40 }
```

Test Input

5
10
20
30

Output

30

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Submitted

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Dequeue
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Prefix Expression
5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct node {
5     int data;
6     struct node *prev;
7     struct node *next;
8 } node;
9
10 node *createnode(int data) {
11     node *newnode = (node *)malloc(sizeof(node));
12     newnode->data = data;
13     newnode->prev = NULL;
14     newnode->next = NULL;
15     return newnode;
16 }
17
18 void insertend(node **head, int data) {
19     node *newnode = createnode(data);
20
21     if (*head == NULL) {
22         *head = newnode;
23         return;
24     }
25
26     node *temp = *head;
27     while (temp->next != NULL) {
28         temp = temp->next;
29     }
30
31     temp->next = newnode;
32     newnode->prev = temp;
33 }
34
35 void deletenode(node **head, node *del) {
36     if (*head == NULL || del == NULL) {
37         return;
38     }
39
40     if (*head == del) {
41         *head = del->next;
42     }
43
44     if (del->prev != NULL) {
45         del->prev->next = del->next;
46     }
47
48     if (del->next != NULL) {
49         del->next->prev = del->prev;
50     }
51
52     free(del);
53 }
54
55 void display(node *head) {
56     node *temp = head;
57     while (temp != NULL) {
58         printf("%d ", temp->data);
59         temp = temp->next;
60     }
61     printf("\n");
62 }
63
64 int main() {
65     node *head = NULL;
66
67     insertend(&head, 10);
68     insertend(&head, 20);
69     insertend(&head, 30);
70     insertend(&head, 40);
71
72     deletenode(&head, head->next); // deletes 20
73
74     display(head);
75
76     return 0;
77 }
```

Test Input

```
4
10 20 30 40
```

Output

```
10 30 40
```

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Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Dequeue
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Suppression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data) {  
9     if (rear == MAX - 1) {  
10         printf("Overflow\n");  
11         return;  
12     }  
13  
14     if (front == -1) {  
15         front = 0;  
16     }  
17  
18     queue[++rear] = data;  
19 }  
20  
21 int dequeue() {  
22     if (front == -1 || front > rear) {  
23         printf("Underflow\n");  
24         return -1;  
25     }  
26  
27     return queue[front++];  
28 }  
29  
30 int main() {  
31     int n, value;  
32  
33     scanf("%d", &n);  
34  
35     for (int i = 0; i < n; i++) {  
36         scanf("%d", &value);  
37         enqueue(value);  
38     }  
39  
40     while (front != -1 || front <= rear) {  
41         printf("%d ", dequeue());  
42     }  
43  
44     return 0;  
45 }
```

Test Input

5
10
20
30

Output

10 20 30 40 50

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Submitted

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertfront operation.

```
void insertfront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 int deque[MAX];  
6 int front = -1, rear = -1;  
7  
8 void insertfront(int data) {  
9     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {  
10         printf("Overflow\n");  
11         return;  
12     }  
13  
14     if (front == -1) {  
15         front = rear = 0;  
16     } else if (front == 0) {  
17         front = MAX - 1;  
18     } else {  
19         front = front - 1;  
20     }  
21  
22     deque[front] = data;  
23 }  
24  
25 void display() {  
26     if (front == -1) {  
27         printf("empty\n");  
28         return;  
29     }  
30  
31     int i = front;  
32     while (1) {  
33         printf("%d ", deque[i]);  
34         if (i == rear) break;  
35         i = (i + 1) % MAX;  
36     }  
37     printf("\n");  
38 }  
39  
40 int main() {  
41     int n, value;  
42  
43     scanf("%d", &n);  
44     for (int i = 0; i < n; i++) {  
45         scanf("%d", &value);  
46         insertfront(value);  
47     }  
48  
49     display();  
50     return 0;  
51 }
```

Test Input

5
10
20
30
..

Output

50 40 30 20 10

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Submitted

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Prefix Supervision
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x) {
4     if (l > r) {
5         return -1;
6     }
7
8     int mid = l + (r - l) / 2;
9
10    if (arr[mid] == x) {
11        return mid;
12    }
13
14    if (arr[mid] > x) {
15        return binarySearch(arr, l, mid - 1, x);
16    }
17
18    return binarySearch(arr, mid + 1, r, x);
19 }
20
21 int main() {
22     int n, x, arr[100];
23
24     scanf("%d", &n);
25
26     for (int i = 0; i < n; i++) {
27         scanf("%d", &arr[i]);
28     }
29
30     scanf("%d", &x);
31
32     int result = binarySearch(arr, 0, n - 1, x);
33
34     printf("%d\n", result);
35
36     return 0;
37 }
```

Test Input

5
10
20
30
...

Output

-1

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QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 void enqueue(int data) {
9     if ((rear + 1) % MAX == front) {
10         printf("Queue full\n");
11         return;
12     }
13
14     if (front == -1) {
15         front = rear = 0;
16     } else {
17         rear = (rear + 1) % MAX;
18     }
19     queue[rear] = data;
20 }
21
22
23 int dequeue() {
24     if (front == -1) {
25         return -1;
26     }
27
28     int data = queue[front];
29
30     if (front == rear) {
31         front = rear = -1;
32     } else {
33         front = (front + 1) % MAX;
34     }
35
36     return data;
37 }
38
39 int main() {
40     int n, x, removed;
41
42     scanf("%d", &n);
43
44     for (int i = 0; i < n; i++) {
45         scanf("%d", &x);
46         enqueue(x);
47     }
48
49     removed = dequeue();
50
51     if (removed == -1) {
52         printf("Queue empty\n");
53     } else {
54         printf("%d\n", removed);
55     }
56
57     return 0;
58 }
```

Submitted

Test Input

5

10

20

30

...

Output

10

Pending manual review

QUESTIONS

Q1

Stack

5 marks

Q2

Circular Queue

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

5 marks

Q10

Stack - Postfix Expression

5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4
5 #define MAX 100
6
7 int stack[MAX];
8 int top = -1;
9
10 void push(int x) {
11     if (top < MAX - 1) {
12         stack[++top] = x;
13     }
14 }
15
16 int pop() {
17     if (top == -1) {
18         return -1;
19     }
20     return stack[top--];
21 }
22
23 int evaluator(char expression[]) {
24     int i = 0;
25
26     while (expression[i] != '\0') {
27         if (isdigit(expression[i])) {
28             push(expression[i] - '0');
29         } else {
30             int op1 = pop();
31             int op2 = pop();
32             int result = 0;
33
34             switch (expression[i]) {
35                 case '+':
36                     result = op1 + op2;
37                     break;
38                 case '-':
39                     result = op1 - op2;
40                     break;
41                 case '*':
42                     result = op1 * op2;
43                     break;
44                 case '/':
45                     result = op1 / op2;
46                     break;
47             }
48             push(result);
49         }
50         i++;
51     }
52     return pop();
53 }
54
55 int main() {
56     char expression[MAX];
57
58     scanf("%s", expression);
59
60     printf("%d\n", evaluator(expression));
61
62     return 0;
63 }
```

Submitted

Test Input

23+

Output

5

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